

Recent Advancements in Industrial Biotechnology

Abstract

Industrial biotechnology, also known as white biotechnology, makes use of living systems, enzymes, and microorganisms to create chemicals, materials, and energy through environmentally friendly methods. With the world dealing with both climate change and the depletion of natural resources, this field has become an important part of the global green economy. This report covers key recent developments in three main areas: biofuels, bioplastics, and industrial enzymes. It looks at how synthetic biology, metabolic engineering, and AI assisted enzyme design are pushing innovation forward, while also discussing ongoing challenges such as scaling up production and managing costs. Overall, the findings highlight the strong potential of industrial biotechnology to drive meaningful change, making it increasingly important to speed up its adoption in the real world.

1. Introduction

The industrial biotechnology worldwide is experiencing significant change due to rising environmental concerns, carbon taxation policies, and an increasing efforts to adopt sustainability. White biotechnology usually applies biological systems to industrial production processes that serves as an environmentally friendly alternative for petroleum-based production. Advanced fermentation processes, cell-free systems, and the development of novel microbial chassis organisms are contributing to the increased commercial viability of white biotechnology products across different sectors including bioenergy, biomaterials, specialty chemicals, and textiles (GlobeNewswire, 2025).

Synthetic biology and artificial intelligence have become an innovative technologies that are reshaping the field. The high demand for alternative proteins, personalized medicine, and sustainable innovative industrial processes played a key role in the significant expansion of . Industrial biotechnology by 2025. Among the most significant advancements in this field are the development of advanced biofuels, biodegradable bioplastics, and highly efficient industrial enzymes. These innovations contribute to reducing greenhouse gas emissions, minimizing waste generation, and improving resource efficiency (Sheldon & Woodley, 2018).

This report provides an in-depth review of recent advancements in three interconnected domains highlighting their key roles, ongoing challenges and future prospects.

2. Biofuels: From First Generation to Fourth Generation

2.1 Evolution of Biofuel Generations

Biofuels are renewable energy sources derived from biological materials such as agricultural residues, algae, plant biomass, and organic waste. They serve as alternatives to conventional fossil fuels and help reduce carbon emissions (Nanda *et al.*, 2022).

Biofuel technology has evolved through distinct generational stages. First-generation (1G) biofuels, derived from food crops such as corn and sugarcane, but have raised food security concerns due to their impact on edible food resources. Second-generation (2G) biofuels, are derived from lignocellulosic biomass such as the fibrous, non-edible parts of plants that offers more safer sustainable alternative. Third-generation (3G) biofuels produced from algae and microalgae as feedstocks, while the emerging fourth-generation (4G) combines carbon capture strategies with fuel production to achieve potentially carbon-negative fuel generation (FEMS Yeast Research, 2025).

Although cellulosic ethanol has demonstrated significant potential as a sustainable biofuel, its commercial deployment has progressed more slowly than initially expected. This slow progress is largely attributed to the expense and technical challenges involved in breaking down the rigid structure of lignocellulosic biomass and efficiently converting the resulting sugars into cost-effective fuels (IEA Bioenergy, 2025).

2.2 Synthetic Biology and Metabolic Engineering

The advancements in synthetic biology and metabolic engineering have significantly transformed the biofuel industry. Scientists have engineered microorganisms such as bacteria, yeast, and algae to improve biomass utilization, yield production under specific industrial conditions. Cellulases, hemicellulases, and ligninases play a crucial role in converting lignocellulosic materials into sugars suitable for fermentation. These enzymes have been optimized to remain active under the high-temperature and acidic conditions commonly encountered during industrial biomass pre-treatment, thereby improving hydrolysis efficiency and reducing the overall cost of biofuel production (RSC Energy Advances, 2025).

The synergistic use of enzyme mixtures such as cellulases, hemicellulases, and ligninases in an optimized proportion has substantially improved the hydrolytic efficiency. Furthermore, CRISPR-Cas systems enable precise genome editing, while de novo pathway engineering recently produces advanced biofuels such as butanol, isoprenoids, and jet fuel substitutes with superior energy density and compatibility with existing fuel infrastructure (ScienceDirect, 2025).

Consolidated bioprocessing (CBP) uses single microorganism for the combined breakdown of biomass and sugar fermentation, reducing the requirement of external enzymes. *Saccharomyces cerevisiae* is widely used because of its well-characterized genome and suitability for genetic modification (PMC, 2025).

2.3 Sustainable Aviation and Marine Fuels

The aviation sector has become a major focus for biofuel innovation. Sustainable aviation fuels (SAFs) derived from biomass, waste oils, and microbial fermentation are increasingly being adopted to reduce carbon emissions from air transport (Nanda *et al.*, 2022). Advances in synthetic biology have enabled microorganisms to produce aviation-compatible hydrocarbons that closely resemble conventional jet fuel (Turner & O'Reilly, 2021). Companies such as LanzaTech are active in converting industrial waste gases into valuable liquid fuels, while Gevo and Aemetis focus on drop-in aviation fuels derived from bio-based feedstocks (GlobeNewswire, 2025).

3. Bioplastics: Scaling Sustainable Polymers

3.1 Market Growth and Environmental Importance

Global plastic production reached 413.8 million metric tonnes in 2023, and more than 90% of these plastics were made from fossil fuels. Although bioplastics currently make up only about 1% of total plastic production, their market is growing quickly. The global biodegradable plastics market was worth around USD 1 billion in 2022 and is expected to increase to USD 2.2 billion by 2028. Most bioplastics are used in flexible and rigid packaging, which account for nearly 60% of their applications (ScienceDirect, 2025). Replacing conventional plastics with bioplastics can provide important environmental benefits. For example, achieving the targets of the European Plastics Strategy could reduce carbon dioxide (CO₂) emissions in Europe by about 13%.

3.2 Polyhydroxyalkanoates (PHAs)

Polyhydroxyalkanoates (PHAs) are biodegradable polyesters produced naturally by microorganisms as internal energy reserves, typically under conditions of environmental stress. Due to their favourable mechanical properties, complete biodegradability, and potential for production from renewable feedstocks, PHAs are increasingly recognised as a leading candidate for next-generation bioplastics (González-Rojo *et al.*, 2024).

Scientists are improving PHA production by changing the internal pathways of microbes so they make more plastic and better-quality plastic. They are also using cheap waste materials, like leftover crops, as food for these microbes. In addition, some microbes have been genetically changed to release the plastic directly into the liquid they grow in, making it easier and cheaper to collect (González-Rojo *et al.*, 2024).

The newest methods combine tough, salt-tolerant microbes called *Halomonas* with AI tools that improve the growing process, along with cleaner methods for processing the final plastic. This approach is helping make large-scale PHA production more practical, bringing the world closer to using these eco-friendly plastics to fight plastic pollution (PMC, 2025). Major companies producing PHA today include Kaneka, Danimer Scientific, BluePHA, and RWDC Industries (Frontiers, 2024).

3.3 Polylactic Acid (PLA) and Other Biopolymers

PLA is made in two steps: microbes first ferment plant sugars into lactic acid, which is then chemically linked together to form the plastic. Top producers include NatureWorks and TotalEnergies Corbion. New fermentation methods are making PLA cheaper to produce, and scientists are creating new PLA blends that are stronger and can handle more heat.

Other biodegradable plastics, like PBS, PBAT, and PBSA, are also becoming popular eco-friendly alternatives. PBS stands out because it can be made partly or fully from plant-based sources (called bio-PBS). New technologies, especially engineered microbes acting as tiny "factories," are making these plastics in cleaner, greener ways (ScienceDirect, 2025).

4. Industrial Enzymes: Engineering the Biocatalytic Revolution

4.1 Market Growth and Expanding Applications

The global market for industrial enzymes was worth about USD 7.4 billion in 2023 and is expected to grow to USD 10.8 billion by 2029, increasing at a rate of 6.5% per year. Food and drinks are still the biggest users of enzymes, but new uses are growing fast in biofuels, bioplastics, cosmetics, medicines, and health supplements. These new uses are creating fresh business opportunities and helping the market grow even more (BCC Research, 2025).

4.2 AI-Driven Enzyme Engineering

AI and machine learning are speeding up enzyme design. A major shift came in January 2024, when Novozymes and Chr. Hansen merged to form Novonesis, which makes enzymes and microbes for food, medicine, household products, bioenergy, and farming while focusing on saving resources (Spherical Insights, 2025). Ginkgo Bioworks is another key player, offering enzyme discovery and organism engineering services across food, pharma, chemicals, and agriculture.

4.3 Lignocellulolytic Enzymes and Biorefineries

Sustainable biorefineries rely on enzymes like cellulases, hemicellulases, and ligninases to break down plant waste into useful products. Scientists are genetically modifying the fungus *Aspergillus niger* to better break down woody biomass (Biotechnology for Biofuels, 2026). These enzyme processes now convert over 80% of raw material, making them cheaper than chemical methods (Nature Communications, 2025). Researchers are improving enzymes to work better in heat and handle tougher materials, while companies like EnginZyme and Solugen are developing "cell-free" enzyme systems for cleaner, more precise chemical production (GlobeNewswire, 2025).

5. Cross-Cutting Themes and Challenges

Across biofuels, bioplastics, and enzymes, similar challenges keep appearing. The biggest issue is scaling up: less than 20% of lab-made microbes succeed in real factories, showing a gap between lab success and real-world use. Making these microbes tough enough to handle factory conditions such as changing raw materials, physical stress, and harmful by-products remains a key challenge (Journal of Biochemical Technology, 2025).

AI and data tools are increasingly used to improve these processes, including virtual models (digital twins), AI-controlled fermentation, and large genetic databases. Companies like Ginkgo Bioworks and Arzeda are turning these tools into real-world services.

Rules and environmental standards are also changing. Carbon taxes, producer responsibility laws, and bio-based content requirements are increasing demand for these products. At the same time, industry groups are creating standard ways to measure eco-friendly products, building more trust and easier comparisons.

6. Conclusion

Industrial biotechnology is growing rapidly due to advances in synthetic biology, metabolic engineering, and artificial intelligence. These technologies are helping replace fossil-based products with sustainable alternatives such as biofuels, bioplastics, and industrial enzymes. Although challenges like cost and large-scale production still exist, continued research, government support, and collaboration will help the industry grow. In the coming years, industrial biotechnology is expected to become an important part of global manufacturing and the green economy.

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